Exactness: DE looks like $\psi_x\,dx + \psi_y\,dy = 0$

for $\psi(y,x) \in C^2$ (twice diff'ble) we have $\psi_{xy} = \psi_{yx}$

this is what we use to check if the DE is exact or not.

For $F(x) = \psi(y(x), x)$

$$\frac{dF}{dx} = \frac{\partial \psi}{\partial y}\frac{dy}{dx} + \frac{\partial \psi}{\partial x} \quad \text{(total derivative)}$$

$F'(x) = 0$, then $F(x) = C$ is a solution for $C \in \mathbb{R}$.

$$\Rightarrow \psi(y(x), x) = C$$

In each of Problems 15 and 16, find the value of b for which the given equation is exact, and then solve it using that value of b.

2.6. 15. $(xy^2 + bx^2y) + (x + y)x^2y' = 0$

Does there exist a $\psi(x,y)$ twice diff'ble, so that

$$\underbrace{(xy^2 + bx^2y)}_{\psi_x} + \underbrace{(x+y)x^2}_{\psi_y}\,y' = \psi_x + \psi_y \cdot y'$$

$$(xy^2 + bx^2y)_y \overset{?}{=} [(x+y)x^2]_x$$

$$2xy + bx^2 \qquad 1 \cdot x^2 + (x+y)2x = \underbrace{x^2 + 2x^2}_{b=3} + 2xy$$

$$\psi_x = xy^2 + 3x^2y \;, \quad \psi_y = x^3 + yx^2$$

$$\psi(y,x) = \frac{x^2}{2}y^2 + x^3y + f(y)$$

$$\psi_y = x^2y + x^3 + f'(y)$$

$f'(y) = 0 \qquad f(y) = \text{const.}$

$$\psi(x,y) = \frac{x^2y^2}{2} + x^3y + c$$

Soln: $\int (xy^2 + bx^2y) + (x+y)x^2y' \, dx = \int 0 \, dx$

$$\frac{x^2y^2}{2} + x^3y + c_1 = c_2 \longrightarrow \text{implicit s.ln.}$$

2.6. 18. Show that any separable equation

$$M(x) + N(y)y' = 0$$

is also exact. exercise

$\mu(x,y)$ "MAKING A DE EXACT"

$$M + Ny' = 0 \qquad \cdot \mu(x,y)$$

$$\underbrace{\mu M}_{\psi_x} + \underbrace{\mu N}_{\psi_y} y' = 0$$

exact $\iff (\mu M)_y = (\mu N)_x$

$$\mu_y M + \mu M_y = \mu_x N + \mu N_x \quad (*)$$

If we assume $\mu = \mu(y)$ $\longrightarrow$ $\begin{cases} \mu_y = \frac{d\mu}{dy} = \mu' \\ \mu_x = 0 \end{cases}$

(*) becomes: $\mu' M + \mu M_y = 0 + \mu N_x$

$$\mu' M = \mu(N_x - M_y)$$

$$\frac{\mu'}{\mu} = \frac{N_x - M_y}{M} = \text{func of } y$$

(do not depend on x $\Leftarrow$ func of y; func of y)

If $\frac{N_x - M_y}{M}$ depends only on y, $\ln|\mu| = \int \frac{N_x - M_y}{M} \, dy$

2.6 24. Show that if $(N_x - M_y)/(xM - yN) = R$, where R depends on the quantity xy only, then the differential equation

$$M + Ny' = 0$$

has an integrating factor of the form $\mu(xy)$. Find a general formula for this integrating factor.

exact $\Leftrightarrow (\mu M)_y = (\mu N)_x$

$$\mu_y M + \mu M_y = \mu_x N + \mu N_x \quad *$$

$\mu(xy)$: Call $xy = v$ $\qquad \mu' = \frac{d\mu}{dv}$, $\frac{\partial v}{\partial x} = y$

$$\mu_x = \frac{\partial \mu(v)}{\partial x} = \frac{d\mu}{dv} \cdot \frac{\partial v}{\partial x}$$

$$\mu_y = \frac{\partial \mu(v)}{\partial y} = \frac{d\mu}{dv} \cdot \frac{\partial v}{\partial y}$$

μ' $\qquad \frac{\partial v}{\partial y} = x$

* becomes:

$$\underbrace{\mu' x}_{=\mu_y} M + \mu M_y = \underbrace{\mu' \cdot y}_{=\mu_x} N + \mu N_x$$

$$\mu'(xM - yN) = \mu(N_x - M_y)$$

$$\frac{\mu'}{\mu} = \frac{N_x - M_y}{xM - yN}$$ } = func of v

func of v ← $v = xy$

sth like $\frac{(xy)^2 + 3xy}{4e^{xy}}$

Assume $\frac{N_x - M_y}{xM - yN}$ is a func of $v = xy$

$$\ln|\mu| = \int \frac{N_x - M_y}{xM - yN}\, dv \quad \Rightarrow \mu = e^{\int \frac{N_x - M_y}{xM - yN} dv}$$

2.6. In each of Problems 25 through 31, find an integrating factor and solve the given equation.

25. $(3x^2y + 2xy + y^2) + (x^2 + y^2)y' = 0$

28. $y + (2xy - e^{-2y})y' = 0$ → exercise

(25) $(3x^2y + 2xy + y^3)_y \overset{?}{=} (x^2+y^2)_x$

$3x^2 + 2x$ $\qquad\qquad$ $= 2x$ $\qquad$ ⇒ not exact.

Let $\mu = \mu(x)$

$$\left[(3x^2y + 2xy + y^3)\mu(x)\right]_y = \left[(x^2+y^2)\mu(x)\right]_x$$

$$(3x^2 + 2x + 3y^2)\mu(x) = (2x)\mu(x) + (x^2+y^2)\mu'(x)$$

$$(3x^2 + \cancel{2x} - \cancel{2x} + 3y^2)\mu(x) = (x^2+y^2)\mu'(x)$$

$$3 = \frac{3x^2+3y^2}{x^2+y^2} = \frac{\mu'}{\mu}$$ ⇒ Then μ is indeed $\mu = \mu(x)$

$$\int 3\,dx = \int \frac{\mu'}{\mu}\,dx$$

$$3x + c = \ln|\mu|$$

$$e^{3x} \cdot C = e^{|\mu|} \quad \rightarrow \mu = e^{3x}$$

$$\underbrace{e^{3x}(3x^2y + 2xy + y^3)}_{\psi_x} + \underbrace{e^{3x}(x^2+y^2)}_{\psi_y} y' = 0$$

$$\psi(x,y) = e^{3x}\left(x^2y + \frac{y^3}{3}\right) + f(x)$$

$$\psi_x = 3e^{3x}\left(x^2y + \frac{y^3}{3}\right) + e^{3x}(2xy) + f'(x)$$

$$\Rightarrow f'(x) = 0 \Rightarrow f(x) = c$$

$$\psi(x,y) = e^{3x}\left(x^2y + \frac{y^3}{3}\right) + c$$

Soln. $(\psi = \text{const})$ $\quad e^{3x}\left(x^2y + \frac{y^3}{3}\right) = \text{const.}$

(28) Hint: $\mu(y) = e^{2y}\cdot y^{-1}$ $\quad$ Soln: $xe^{2y} - \ln y = c$

Misc. Probs.

33. **Riccati Equations.** The equation

$$\frac{dy}{dt} = q_1(t) + q_2(t)y + q_3(t)y^2$$

is known as a Riccati[11] equation. Suppose that some particular solution y_1 of this equation is known. A more general solution containing one arbitrary constant can be obtained through the substitution

$$y = y_1(t) + \frac{1}{v(t)}$$

Show that $v(t)$ satisfies the first order *linear* equation

$$\frac{dv}{dt} = -(q_2 + 2q_3y_1)v - q_3.$$

Note that $v(t)$ will contain a single arbitrary constant.

y_1 is a soln:

$$y_1' = q_1 + q_2y_1 + q_3y_1^2$$

$$y = y_1(t) + \frac{1}{v(t)} \xrightarrow{\text{diff.}} y' = y_1' - \frac{v'}{v^2}$$

$$y^2 + 2\frac{y_1}{v} + \frac{1}{v^2}$$

$$y_1' - \frac{v'}{v^2} = q_1 + q_2\left(y_1 + \frac{1}{v}\right) + q_3\left(y_1 + \frac{1}{v}\right)^2$$

$$= q_1 + q_2y_1 + \frac{q_2}{v} + q_3y_1^2 + 2q_3\frac{y_1}{v} + \frac{q_3}{v^2}$$

$$-\frac{v'}{v^2} = \frac{q_2}{v} + 2q_3\frac{y_1}{v} + \frac{q_3}{v^2} \qquad (\times v^2)$$

$$v' = -(q_2 + 2q_3y_1)v - q_3$$

Misc. Prob.

34. Using the method of Problem 33 and the given particular solution, solve each of the following Riccati equations:

(a) $y' = \underbrace{1}_{q_1} + \underbrace{t^2}_{q_2} \underbrace{-2ty}_{q_3} + y^2$; $\quad y_1(t) = t$

$$y = y_1 + \frac{1}{v}, \qquad y_1 = t$$

$$v' = -(-2t + 2t)v - 1 \quad \Rightarrow \quad v' = -1$$

($-2t+2t = 0$)

$$\Rightarrow \quad v = -t + C$$

$$\Rightarrow \quad y = t + \frac{1}{C - t}$$

1st order DE:

- linear $\longrightarrow$ $\mu(x)$ integrating factor
- nonlinear:
  - separable $\longrightarrow$ $\int A(x)\,dx = \int B(y)\,dy$
  - exact $\longrightarrow$ Find ψ s.t. $\psi_x\,dx + \psi_y\,dy = 0$

    soln: $\psi = \text{constant}$.
  - not exact $\longrightarrow$ Make the DE exact & solve.

    { Is there $\mu(x,y)$ s.t. $\mu \cdot (\text{DE})$ is exact? }
  - nothing works: Is it homogeneous?

$$y' = F(x,y) \overset{?}{=} F\left(\frac{y}{x}\right) \qquad \text{if true } = F\left(\frac{\lambda y}{\lambda x}\right)$$

check: $F(x,y) \overset{?}{=} F(\lambda x, \lambda y)$

If homogeneous, $v = \frac{y}{x} \Rightarrow$ DE (4) is separable. Solve.

In each of Problems 1 and 2, transform the given initial value problem into an equivalent problem with the initial point at the origin.

2.8. 1. $dy/dt = t^2 + y^2$, $y(1) = 2$

$t \longmapsto s$, $y \longmapsto z$ s.t. $z(0) = 0$

$s+1 = t$

If we let $s = t-1$ and define $z = y-2$. Then

$s = 0 \Rightarrow t = 1 \Rightarrow y = 2 \Rightarrow z = 0$.

Now we write the DE $z(s)$ satisfies:

$$\frac{dz(s)}{ds} = \frac{dz(t(s))}{dt}\,\frac{dt(s)}{ds} = \frac{d(y-2)}{dt}\cdot 1 = \frac{dy}{dt} = t^2 + y^2$$

$$= (s+1)^2 + (z+2)^2$$

So we have

$$z' = \frac{dz}{ds} = (s+1)^2 + (z+2)^2 \quad \text{with IV: } z(0) = 0.$$

In each of Problems 3 through 6, let $\phi_0(t) = 0$ and define $\{\phi_n(t)\}$ by the method of successive approximations

(a) Determine $\phi_n(t)$ for an arbitrary value of n.

~~(b) Plot $\phi_n(t)$ for $n = 1, \ldots, 4$. Observe whether the iterates appear to be converging.~~

(c) Express $\lim_{n\to\infty} \phi_n(t) = \phi(t)$ in terms of elementary functions; that is, solve the given initial value problem.

2.8. 3. $y' = 2(y+1)$, $y(0) = 0$

(a) We'll use $\phi_{n+1}(t) = \int_0^t 2(\phi_n(s)+1)\,ds$.

Start with $\phi_0(t) = 0$, $\phi_1(t) = \int_0^t 2(\phi_0(s)+1)\,ds = \int_0^t 2\,ds = [2s]_{s=0}^t = 2t$

$$\phi_2(t) = \int_0^t 2(\phi_1(s)+1)\,ds = \int_0^t 4s+2\,ds = [2s^2+2s]_{s=0}^t = \underbrace{2t^2+2t}_{\phi_1}$$

$$\phi_3(t) = \int_0^t 2(\phi_2(s)+1)\,ds = \int_0^t 4s^2+4s+2\,ds = \left[\tfrac{4}{3}s^3+2s^2+2s\right]_{s=0}^t = \frac{4t^3}{3} + \underbrace{2t^2+2t}_{\phi_2}$$

Note that $2t^2 = \int_0^t 2\cdot(2s)\,ds$ and $\frac{4t^3}{3} = \int_0^t 2\cdot(2s^2)\,ds$

So for ϕ_4, we will add $\int_0^t 2 \cdot \frac{4s^3}{3}\,ds$ to ϕ_3, and so on:

$\phi_1 = 2t$

$\phi_2 = 2 \cdot t^2 + \phi_1$

$\phi_3 = 2 \cdot 2 \cdot \frac{t^3}{3} + \phi_2$

$\phi_4 = 2 \cdot 2 \cdot 2 \cdot \frac{t^4}{3 \cdot 4} + \phi_3$

$\phi_5 = 2 \cdot 2 \cdot 2 \cdot 2 \cdot \frac{t^5}{3 \cdot 4 \cdot 5} + \phi_4 = 2^5 \cdot \frac{t^5}{2 \cdot 3 \cdot 4 \cdot 5} + \phi_4 = 2^5 \frac{t^5}{5!} + \phi_4$

$\vdots$

$\phi_k = 2^k \frac{t^k}{k!} + \phi_{k-1}$ ($\phi_2 - \phi_1$, ϕ_1)

$= 2^k \frac{t^k}{k!} + 2^{k-1} \frac{t^{k-1}}{(k-1)!} + 2^{k-2} \frac{t^{k-2}}{(k-2)!} + \cdots + 2^2 \frac{t^2}{2} + 2 \cdot \frac{t^1}{1!}$ ($i=k-2$, $i=2$, $i=1$)

$= \sum_{i=1}^{k} \frac{2^i t^i}{i!}$

You can use mathematical induction to prove it.

- Basis step: ϕ_1 ✓
- Inductive step: Assume $\phi_k = \sum_{i=1}^{k} \frac{2^i t^i}{i!}$, show $\phi_{k+1} = \sum_{i=1}^{k+1} \frac{2^i t^i}{i!}$ (∫... =?)

$$\phi_{k+1} = \int_0^t 2(\phi_k(s)+1)\,ds = \int_0^t 2 \sum_{i=1}^{k} \frac{2^i s^i}{i!} + 2\,ds = \int_0^t \sum_{i=1}^{k} \frac{2^{i+1} s^i}{i!}\,ds + 2t$$

$$= \sum_{i=1}^{k} \int_0^t \frac{2^{i+1} s^i}{i!}\,ds + 2t = \sum_{i=1}^{k} \frac{2^{i+1}}{i!} \cdot \left[\frac{s^{i+1}}{(i+1)}\right]_{s=0}^{t} + 2t = \sum_{i=1}^{k} \frac{2^{i+1} t^{i+1}}{(i+1)!} + 2t$$

$(j = i+1)$

$$= \sum_{j=2}^{k+1} \frac{2^j \cdot t^j}{j!} + 2t$$

$$= \sum_{j=1}^{k+1} \frac{2^j \cdot t^j}{j!}$$

(c) $\lim_{n\to\infty} \sum_{i=1}^{n} \frac{(2t)^i}{i!} + 1 - 1 = \lim_{n\to\infty} \sum_{i=0}^{n} \frac{(2t)^n}{n!} - 1$

$= e^{2t} - 1$

In each of Problems 1 through 8, find the general solution of the given differential equation.

3.1. 2. $y'' + 3y' + 2y = 0$

2nd order, linear, ODE
hom. eqn. w/ const. coef.

$y'' - y = 0$ $\quad e^t$, e^{-t}
$y'' = y$

$y = e^{rt}$, $y' = re^{rt}$, $y'' = r^2 e^{rt}$

$y'' + ay' + by = 0 \Rightarrow \underbrace{(r^2 + ar + b)}_{\text{char. eqn.} = 0} \underbrace{e^{rt}}_{\neq 0} = 0$

y_1 soln. $\Rightarrow cy_1$ soln.

$c_1 y_1 + c_2 y_2$

$(cy_1)' = cy_1'$ $\quad (cy_1)'' = cy_1''$ $\quad c_1(y_1'' + ay_1' + by_1) = 0$
$+ c_2(y_2'' \cdots)$

char. eqn: $r^2 + 3r + 2 = 0 \Rightarrow r_1 = -2 \quad r_2 = -1$
$(r+1)(r+2)$

$y_1 = c_1 e^{-2t} \quad y_2 = c_2 e^{-t}$

general soln: $y = c_1 e^{-2t} + c_2 e^{-t}$

In each of Problems 9 through 16, find the solution of the given initial value problem. Sketch the graph of the solution and describe its behavior as t increases.

3.1. 10. $y'' + 4y' + 3y = 0, \quad y(0) = 2, \quad y'(0) = -1$

char. eqn: $r^2 + 4r + 3 = 0 \Rightarrow r_1 = -1 \quad r_2 = -3$

$y_1 = c_1 e^{-t} \quad y_2 = c_2 e^{-3t}$

general soln: $y = c_1 e^{-t} + c_2 e^{-3t} \rightarrow y' = -c_1 e^{-t} - 3c_2 e^{-3t}$

IVs: $y(0) = 2 \Rightarrow c_1 + c_2 = 2$
$y'(0) = -1 \Rightarrow -c_1 - 3c_2 = -1$

$-2c_2 = 1$
$c_2 = -1/2 \quad c_1 = 5/2$

soln to the IVP: $y = \frac{5}{2}e^{-t} - \frac{1}{2}e^{-3t}$

3.1.17. Find a differential equation whose general solution is $y = c_1e^{2t} + c_2e^{-3t}$.

char eqn: $(r-2)(r+3) = 0$ $\quad r_1 = 2, \; r_2 = -3$

$r^2 + r - 6 = 0$

DE: $y'' + y' - 6y = 0$

3.1.21. Solve the initial value problem $y'' - y' - 2y = 0$, $y(0) = \alpha$, $y'(0) = 2$. Then find α so that the solution approaches zero as $t \to \infty$.

char eqn: $r^2 - r - 2 = 0$ $\quad (r-2)(r+1)$ $\quad \Rightarrow r_1 = 2, \; r_2 = -1$

$y_1 = c_1e^{2t}$ $\quad y_2 = c_2e^{-t}$

general soln: $y = c_1e^{2t} + c_2e^{-t}$

$y' = 2c_1e^{2t} - c_2e^{-t}$

IVs: $y(0) = \alpha \Rightarrow c_1 + c_2 = \alpha$

$y'(0) = 2 \Rightarrow 2c_1 - c_2 = 2$

$3c_1 = \alpha + 2$

$c_1 = \frac{\alpha+2}{3}$

$c_2 = \frac{2\alpha-2}{3}$

soln: $y = \frac{\alpha+2}{3}e^{2t} + \frac{2\alpha-2}{3}e^{-t}$

$\frac{\alpha+2}{3}e^{2t} \to \alpha$ as $t \to \infty$; $\frac{2\alpha-2}{3}e^{-t} \to 0$ as $t \to \infty$

$\frac{\alpha+2}{3} = 0 \quad \alpha = -2$